

Candidate additions to the introduction of *Bayesian nonparametric fusion of randomised and real-world survival data for heterogeneous treatment effects*

Working note

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This note inventories papers that could be added to the two literature paragraphs of the introduction (the confounding-function / frequentist paragraph and the Bayesian-borrowing paragraph). Organisation is by importance: the single closest competitor that is currently missing; references already in the working-paper bibliography that could be promoted to the introduction; references not yet on file that fit the narrative. BibTeX entries for every paper are embedded at the top of this file (the `filecontents` block writes `intro_additions.bib` on first compile).

1. The single most important omission

Yozova et al. [2025] extend Bayesian Causal Forests to fuse trial and confounded observational data through an additively augmented BCF model with source-specific prognostic and treatment-effect components and a data-driven tempering of the observational likelihood. This is the closest Bayesian competitor to the proposed method: Bayesian, tree-based, fuses an RCT with a confounded observational source, and explicitly models the bias from unmeasured confounding. The differences are the outcome type (continuous rather than survival) and the bias-correction mechanism (tempered likelihood rather than a confounding-function decomposition under an AFT). A reasonable insertion in Paragraph 2 is: “*Yozova et al. [2025] extend BCF to fuse trial and confounded observational data through a tempered-likelihood construction, but for continuous outcomes.*” The closing gap statement of Paragraph 2 sharpens once this paper is acknowledged explicitly.

2. Already in the working-paper references; candidates for promotion to the introduction

Paragraph 1 (frequentist).

- Cheng and Cai [2021] propose an adaptive combination of RCT and observational data for the ATE, with a data-driven weight that interpolates between trial-only and pooled estimators. Natural ATE precursor before the HTE-targeted Yang line.
- Shyr et al. [2025] develop a multi-study R-learner for HTE estimation across multiple studies under within-study unconfoundedness. Belongs next to ? and broadens the multi-source HTE strand.
- Yang and Ding [2020] construct a unified estimating-equation framework for combining multiple observational datasets, optimally weighting sources to minimise asymptotic variance. A more general data-fusion precursor.

- van der Laan et al. [2024] develop an adaptive targeted maximum-likelihood estimator that fuses RCT and RWD with a data-driven RWD-influence mechanism. Rounds out the frequentist ATE-fusion landscape.
- Lin et al. [2024] critically compare the confounding-function approach with experimental grounding under simulations for ATE estimation. Useful both as a comparative reference and because their simulations question whether either approach reliably beats RCT-only for ATE—which makes the HTE focus of the present work more pointed.

Paragraph 2 (Bayesian).

- Müller et al. [2023] review Bayesian methodologies for integrating real-world data into clinical studies, covering power priors, MAP, commensurate priors and recent developments. Useful as the Bayesian counterpart of Colnet et al. [2024]: a one-stop “see X for review” citation that avoids enumerating every variant.
- Murray et al. [2015] extend commensurate priors to nonexchangeable functional and survival sources in oncology, using generalised mixture commensurate priors. Fits alongside ? as a Bayesian-borrowing precursor for survival data.
- Ohigashi et al. [2022] and Ohigashi et al. [2025] develop horseshoe and other continuous-shrinkage priors for dynamic borrowing of multiple historical controls. Optional additions to the “later refinements” lineage.

3. Already in the methodology references; candidates for promotion

- Henderson et al. [2020] develop a fully nonparametric Bayesian accelerated failure time model for individualised treatment effects with censored data, in a single source under unconfoundedness. This is the natural single-source Bayesian-AFT HTE precursor to the proposed method and is currently cited only in the methodology section. Promoting it to the introduction (last sentence of Paragraph 2, before the gap statement) tightens the gap argument: “*Henderson et al. [2020] develop nonparametric Bayesian AFT modelling for individualised treatment effects in a single source.*”
- Hahn et al. [2020] introduce Bayesian Causal Forests with a confounding-aware decomposition into prognostic and treatment-effect surfaces. Already cited in the methodology; a brief mention in the introduction would place the BCF lineage explicitly.

4. Not currently in the references; plausible additions

- Hu et al. [2021] compare Bayesian AFT-BART and other machine-learning methods for HTE estimation in observational survival data under unconfoundedness, finding that AFT-BART performs best on bias, precision and credible-interval coverage. The closest single-source Bayesian-survival-HTE precursor to the proposed method.
- Zhao et al. [2025] propose a doubly robust transfer-learning framework for the optimal individualised treatment regime under right censoring, combining trial and observational data with covariate shift. A related but distinct estimand (decision rule rather than CATE) and bias mechanism (covariate shift rather than unmeasured confounding); worth one sentence to delineate.

- Colnet et al. [2024] provide a comprehensive review of frequentist data-fusion methods. Saves listing every variant on the frequentist side.
- Bareinboim and Pearl [2016] establish the theory of identifiability for data-fusion problems. Useful at the start of Paragraph 1 if a foundational reference is desired.
- Bi et al. [2023] construct a Bayesian nonparametric hybrid control (PAM-HC) for randomised clinical trials using external data. Pairs with ? as a Bayesian-BART precursor.
- Suggested but with BibTeX details still to confirm: Lee et al. (2022, 2023, 2024) on doubly robust generalisability for survival; Alt et al. (2023) LEAP prior for dynamic Bayesian borrowing; Wang & Rosner (2019) on Bayesian nonparametric multi-source propensity-score adjustment.

5. Test-then-pool and selective-borrowing strand

A parallel strand to the confounding-function modelling addresses RWD bias by selection rather than by estimation. The canonical anchors are ? for test-then-pool, ? for weighted combination, and ? for adaptive selection. A single sentence in Paragraph 1 with two or three of these citations is sufficient to delineate the contrast with the confounding-function modelling approach taken in this work. (BibTeX entries to add once verified.)

Minimal recommendation

If only three papers are added, the following are the highest-impact:

1. **Yozova et al. [2025]** — closest Bayesian competitor; currently missing.
2. **Henderson et al. [2020]** — natural single-source Bayesian-AFT HTE precursor; sharpens the final gap statement.
3. **Colnet et al. [2024]** or **Lin et al. [2024]** — umbrella review citation, replaces enumeration on the frequentist side.

A slightly expanded version adds Shyr et al. [2025] to extend the HTE-fusion lineage and Cheng and Cai [2021] to cover the adaptive-combination ATE precursor.

References

- Elias Bareinboim and Judea Pearl. Causal inference and the data-fusion problem. *Proceedings of the National Academy of Sciences*, 113(27):7345–7352, 2016.
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- Liangyuan Hu, Jiayi Ji, and Fan Li. Estimating heterogeneous survival treatment effect in observational data using machine learning. *Statistics in Medicine*, 40(21):4691–4713, 2021. doi: 10.1002/sim.9090.
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- Peter Müller, Noirrit Kiran Chandra, and Abhra Sarkar. Bayesian approaches to include real-world data in clinical studies. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 381(2247):20220158, 2023. doi: 10.1098/rsta.2022.0158.
- Timothy A. Murray, Brian P. Hobbs, and Bradley P. Carlin. Combining nonexchangeable functional or survival data sources in oncology using generalized mixture commensurate priors. *The Annals of Applied Statistics*, 9(3):1549–1570, 2015.
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- Tetsuhisa Ohigashi, Hisato Ochi, Shintaro Aoki, and Chikuma Hamada. Potential bias models with Bayesian shrinkage priors for dynamic borrowing of multiple historical control data. *Pharmaceutical Statistics*, 24(2):e2453, 2025.
- Cathy Shyr, Bo Ren, Prasad Patil, and Giovanni Parmigiani. Multi-study R-learner for estimating heterogeneous treatment effects across studies using statistical machine learning. *Biostatistics*, 26(1):kxaf040, 2025.
- Mark J. van der Laan, Shuangning Qiu, and Lisha van der Laan. Adaptive-TMLE for the average treatment effect based on randomized controlled trial augmented with real-world data. *arXiv preprint arXiv:2405.07186*, 2024.
- Shu Yang and Peng Ding. Combining multiple observational data sources to estimate causal effects. *Journal of the American Statistical Association*, 115(531):1540–1554, 2020. doi: 10.1080/01621459.2019.1609973.
- Irina Yozova, Nathan McJames, and Ioanna Manolopoulou. Data fusion under weak identifiability in heterogeneous treatment effect modelling, 2025. O’Bayes Conference on Objective Bayes Methodology, Athens, Greece, June 2025. Conference presentation.
- Pan Zhao, Julie Josse, and Shu Yang. Efficient and robust transfer learning of optimal individualized treatment regimes with right-censored survival data. *Journal of Machine Learning Research*, 26, 2025. arXiv:2301.05491.